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Department of Mathematics

LECTURE NOTES

Advanced Calculus and Numerical Methods (21MAT21)

Module 2 (Vector Calculus)

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Module - 2 - Vector Calculus

Vectors differentiation:-

1. Let $P(x, y, z)$ be any point in XYZ plane, then the position vector P can be defined as $x\bar{i} + y\bar{j} + z\bar{k}$.

Its derivative becomes $d\vec{r} = dx\bar{i} + dy\bar{j} + dz\bar{k}$
and suppose x, y, z are the function in variable 't'
then $\vec{r} = x(t)\bar{i} + y(t)\bar{j} + z(t)\bar{k}$ and its derivative is

$$\frac{d\vec{r}}{dt} = \frac{dx}{dt}\bar{i} + \frac{dy}{dt}\bar{j} + \frac{dz}{dt}\bar{k}.$$

2. Let $\vec{A} = a_1\bar{i} + a_2\bar{j} + a_3\bar{k}$, $\vec{B} = b_1\bar{i} + b_2\bar{j} + b_3\bar{k}$ can represent the two surfaces, then

$\vec{A} \cdot \vec{B} = a_1b_1 + a_2b_2 + a_3b_3$ is a scalar

$$\vec{A} \times \vec{B} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = c_1\bar{i} + c_2\bar{j} + c_3\bar{k}$$

represents a vector point function.

$\Rightarrow \cos \theta = \frac{\vec{A} \cdot \vec{B}}{|\vec{A}| \cdot |\vec{B}|}$, where ' θ ' is the angle between the given surfaces, $|\vec{A}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$ and

$$|\vec{B}| = \sqrt{b_1^2 + b_2^2 + b_3^2}.$$

3) Gradient:- (∇ or grad)

Let $\vec{i}, \vec{j}, \vec{k}$ be the vectors can represent $\vec{OX}, \vec{OY}, \vec{OZ}$ then the gradient is defined as

$$\nabla = \text{grad} = \frac{\partial}{\partial x} \vec{i} + \frac{\partial}{\partial y} \vec{j} + \frac{\partial}{\partial z} \vec{k} \text{ be a vector point function.}$$

4) $\phi(x, y, z) = C$ be a scalar function, then the gradient of ϕ can be defined as

$$\nabla \phi = \text{grad} \phi = \left(\frac{\partial}{\partial x} \vec{i} + \frac{\partial}{\partial y} \vec{j} + \frac{\partial}{\partial z} \vec{k} \right) \phi = \frac{\partial \phi}{\partial x} \vec{i} + \frac{\partial \phi}{\partial y} \vec{j} + \frac{\partial \phi}{\partial z} \vec{k}.$$

5) Let $\phi_1(x, y, z) = C_1$, $\phi_2(x, y, z) = C_2$ be the two surfaces intersecting at the point 'p' with the angle ' θ ', then

$$\cos \theta = \frac{\nabla \phi_1 \cdot \nabla \phi_2}{|\nabla \phi_1| |\nabla \phi_2|}$$

if the given surfaces can intersect orthogonally then

$$\nabla \phi_1 \cdot \nabla \phi_2 = 0.$$

6) Let $\vec{F} = F_1 \vec{i} + F_2 \vec{j} + F_3 \vec{k}$ where $F_1 = F_1(x, y, z)$,

$F_2 = F_2(x, y, z)$, $F_3 = F_3(x, y, z)$ be a vector point function then the Curl of $\vec{F}$ is defined as

$$\text{Curl } \vec{F} = \nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix}$$

can represent a vector point function.

7) Let $F = F_1\vec{i} + F_2\vec{j} + F_3\vec{k}$, where F_1, F_2, F_3 are the functions in x, y, z then

$$\nabla \cdot \vec{F} = \left(\frac{\partial}{\partial x} \vec{i} + \frac{\partial}{\partial y} \vec{j} + \frac{\partial}{\partial z} \vec{k} \right) (F_1\vec{i} + F_2\vec{j} + F_3\vec{k}) = \text{div. } \vec{F} \quad (\text{divergent})$$

$$\Rightarrow \nabla \cdot \vec{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} \text{ is a scalar.}$$

if $\nabla \cdot \vec{F} = 0$, then the given vector point function $\vec{F}$ is called "solenoidal".

if $\nabla \times \vec{F} = 0\vec{i} + 0\vec{j} + 0\vec{k} = \vec{0}$, then $\vec{F}$ is known as "irrotational".

8) Let $\phi(x, y, z) = C$ be a scalar function and $\vec{d}$ be a direction vector, then the directional derivative can be defined as $\nabla \phi \cdot \hat{n}$.

$$\hat{n} = \frac{\vec{d}}{|\vec{d}|} \quad (\text{normal})$$

9) Let $\vec{F} = F_1\vec{i} + F_2\vec{j} + F_3\vec{k}$ be a vector point function, let $\phi(x, y, z)$ be a scalar potential or conservative force field is defined as

$$\vec{F} = \nabla \phi.$$

1. Find the angle between the surfaces $x^2 + y^2 + z^2 = 9$ and $z = x^2 + y^2 - 3$ at a point $(2, -1, 2)$.

Sol:- Given,

$$\phi_1 = x^2 + y^2 + z^2 = 9 \rightarrow (1)$$

$$\phi_2 = z = x^2 + y^2 - 3$$

$$\Rightarrow \phi_2 = x^2 + y^2 - z = 3 \rightarrow (2)$$

and $P(2, -1, 2)$

$$\therefore \nabla \phi_1 = \frac{\partial \phi_1}{\partial x} \bar{i} + \frac{\partial \phi_1}{\partial y} \bar{j} + \frac{\partial \phi_1}{\partial z} \bar{k}$$

$$\Rightarrow \nabla \phi_1 = 2x\bar{i} + 2y\bar{j} + 2z\bar{k}$$

$$\therefore (\nabla \phi_1)_P = 2(2)\bar{i} + 2(-1)\bar{j} + 2(2)\bar{k}$$

$$\Rightarrow (\nabla \phi_1)_P = 4\bar{i} - 2\bar{j} + 4\bar{k} = \vec{A}$$

$$\therefore |\vec{A}| = \sqrt{(4)^2 + (-2)^2 + (4)^2}$$

$$\Rightarrow |\vec{A}| = \sqrt{16 + 4 + 16}$$

$$\Rightarrow |\vec{A}| = \sqrt{36}$$

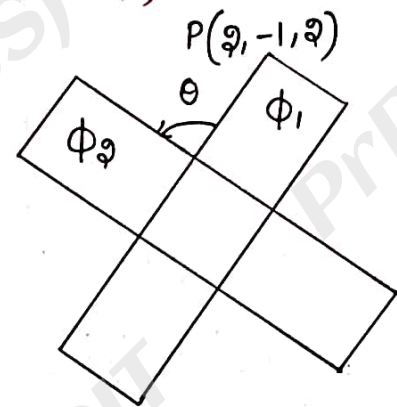
$$\Rightarrow |\vec{A}| = 6$$

$$\therefore \nabla \phi_2 = \frac{\partial \phi_2}{\partial x} \bar{i} + \frac{\partial \phi_2}{\partial y} \bar{j} + \frac{\partial \phi_2}{\partial z} \bar{k}$$

$$\Rightarrow \nabla \phi_2 = 2x\bar{i} + 2y\bar{j} - \bar{k}$$

$$\therefore (\nabla \phi_2)_P = 2(2)\bar{i} + 2(-1)\bar{j} - \bar{k}$$

$$\Rightarrow (\nabla \phi_2)_P = 4\bar{i} - 2\bar{j} - \bar{k} = \vec{B}$$



$$\therefore |\vec{B}| = \sqrt{(4)^2 + (-2)^2 + (-1)^2}$$

$$\Rightarrow |\vec{B}| = \sqrt{16+4+1}$$

$$\Rightarrow |\vec{B}| = \sqrt{21}$$

$\therefore$ WKT,

$$\cos \theta = \frac{\vec{A} \cdot \vec{B}}{|\vec{A}| |\vec{B}|} = \frac{(4\vec{i} - 2\vec{j} + 4\vec{k}) (4\vec{i} - 2\vec{j} - \vec{k})}{6\sqrt{21}}$$

$$\Rightarrow \cos \theta = \frac{16+4-4}{6\sqrt{21}} = \frac{16}{3\sqrt{21}}$$

$$\Rightarrow \theta = \cos^{-1}\left(\frac{8}{3\sqrt{21}}\right)$$

Q. Find the angle between the surfaces $x^2 + y^2 - z^2 = 4$,
 $x^2 + y^2 - 30 = z$ at the point $(2, 1, 2)$.

Sol:- Given,

$$\phi_1 = x^2 + y^2 - z^2 = 4 \rightarrow (1)$$

$$\phi_2 = x^2 + y^2 - z = 30 \rightarrow (2)$$

and $P = (2, 1, 2)$

$$\therefore \nabla \phi_1 = \frac{\partial \phi_1}{\partial x} \vec{i} + \frac{\partial \phi_1}{\partial y} \vec{j} + \frac{\partial \phi_1}{\partial z} \vec{k}$$

$$\Rightarrow \nabla \phi_1 = 2x\vec{i} + 2y\vec{j} - 2z\vec{k}$$

$$\therefore (\nabla \phi_1)_P = 2(2)\vec{i} + 2(1)\vec{j} - 2(2)\vec{k}$$

$$\Rightarrow (\nabla \phi_1)_P = 4\vec{i} + 2\vec{j} - 4\vec{k}$$

$$\therefore |\vec{A}| = \sqrt{(4)^2 + (2)^2 + (-4)^2}$$

$$\Rightarrow |\vec{A}| = \sqrt{16+4+16}$$

$$\Rightarrow |\vec{A}| = \sqrt{36}$$

$$\Rightarrow |\vec{A}| = 6$$

$$\therefore \nabla \phi = \frac{\partial \phi}{\partial x} \vec{i} + \frac{\partial \phi}{\partial y} \vec{j} + \frac{\partial \phi}{\partial z} \vec{k}$$

$$\Rightarrow \nabla \phi = 4x\vec{i} + 2y\vec{j} - \vec{k}$$

$$\therefore (\nabla \phi)_P = 4(2)\vec{i} + 2(1)\vec{j} - \vec{k}$$

$$\Rightarrow (\nabla \phi)_P = 4\vec{i} + 2\vec{j} - \vec{k}$$

$$\therefore |\vec{B}| = \sqrt{(4)^2 + (2)^2 + (-1)^2}$$

$$\Rightarrow |\vec{B}| = \sqrt{16+4+1}$$

$$\Rightarrow |\vec{B}| = \sqrt{21}$$

$$\therefore \text{WKT, } \cos \theta = \frac{\vec{A} \cdot \vec{B}}{|\vec{A}| \cdot |\vec{B}|} = \frac{(4\vec{i} + 2\vec{j} - \vec{k}) \cdot (4\vec{i} + 2\vec{j} - \vec{k})}{6\sqrt{21}}$$

$$\Rightarrow \cos \theta = \frac{16+4+4}{6\sqrt{21}}$$

$$\Rightarrow \cos \theta = \frac{24}{6\sqrt{21}}$$

$$\Rightarrow \cos \theta = \frac{4}{\sqrt{21}}$$

$$\Rightarrow \theta = \cos^{-1}\left(\frac{4}{\sqrt{21}}\right)$$

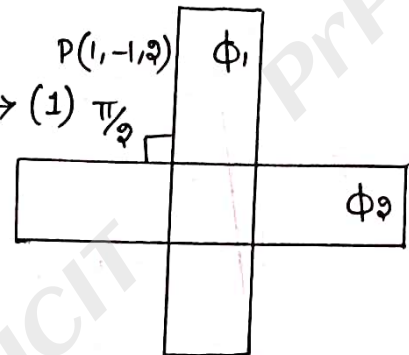
3. Find the value of a and b , such that the surfaces $ax^2 + byz = (a+b)x$ and $4x^2y + z^3 = 4$ are orthogonal at the point $(1, -1, 2)$.

Sol:- Given,

$$\phi_1 = ax^2 - byz = (a+b)x \rightarrow (1) \quad \pi/2$$

$$\phi_2 = 4x^2y + z^3 = 4 \rightarrow (2)$$

and $P = (1, -1, 2)$



WKT,

The point $P(1, -1, 2)$ lies on ϕ_1 & ϕ_2 .

$$\therefore (1) \Rightarrow a(1)^2 - b(-1)(2) = (a+b)(1)$$

$$\Rightarrow a + 2b = a + b$$

$$\Rightarrow 2b = b$$

$$\Rightarrow b = 1$$

$$\therefore (1) \Rightarrow \phi_1 = ax^2 - yz - (a+b)x = 0 \rightarrow (3)$$

$$\therefore \nabla \phi_1 = \frac{\partial \phi_1}{\partial x} \bar{i} + \frac{\partial \phi_1}{\partial y} \bar{j} + \frac{\partial \phi_1}{\partial z} \bar{k}$$

$$\Rightarrow \nabla \phi_1 = (2ax - a - b)\bar{i} - z\bar{j} - y\bar{k}$$

$$\therefore (\nabla \phi_1)_P = (a-2)\bar{i} - 2\bar{j} + \bar{k}$$

$$\therefore \nabla \phi_2 = \frac{\partial \phi_2}{\partial x} \bar{i} + \frac{\partial \phi_2}{\partial y} \bar{j} + \frac{\partial \phi_2}{\partial z} \bar{k}$$

$$\Rightarrow \nabla \phi_2 = 8xy\bar{i} + 4x^2\bar{j} + 3z^2\bar{k}$$

$$\therefore (\nabla \phi_2)_P = -8\bar{i} + 4\bar{j} + 12\bar{k}$$

But given ϕ_1 & ϕ_2 can intersect orthogonally,

$$\nabla\phi_1 \cdot \nabla\phi_2 = 0$$

$$|(a-2)\bar{i} - 2\bar{j} + \bar{k}| \cdot |-8\bar{i} + 4\bar{j} + 12\bar{k}| = 0$$

$$\Rightarrow -8(a-2) - 8 + 12 = 0$$

$$\Rightarrow -8a - 16 - 8 + 12 = 0$$

$$\Rightarrow 20 - 8a = 0$$

$$\Rightarrow 20 = 8a$$

$$\Rightarrow a = \frac{20}{8}$$

$$\Rightarrow \boxed{a = \frac{5}{2}}$$

4. Find the angle between the normal to the surface $xy = z^2$ at the point $(4, 1, 2)$ and $(3, 3, -3)$.

Sol: Given,

$$\phi_1 = xy = z^2$$

$$\Rightarrow \phi_1 = xy - z^2 = 0$$

$$\therefore \nabla\phi = \frac{\partial\phi}{\partial x}\bar{i} + \frac{\partial\phi}{\partial y}\bar{j} + \frac{\partial\phi}{\partial z}\bar{k}$$

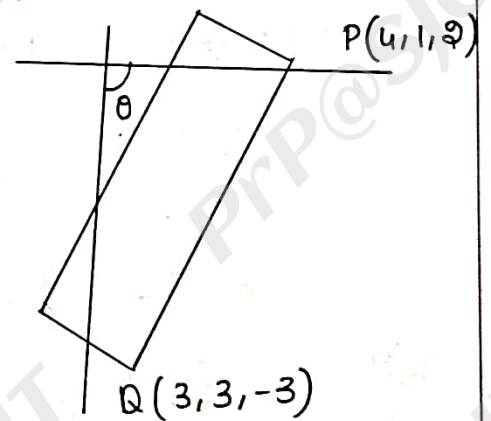
$$\Rightarrow \nabla\phi = y\bar{i} + x\bar{j} - 2z\bar{k}$$

$$\therefore (\nabla\phi)_{P(4,1,2)} = \bar{i} + 4\bar{j} - 4\bar{k} = \vec{A} \text{ (say)}$$

$$(\nabla\phi)_{P(3,3,-3)} = 3\bar{i} + 3\bar{j} + 6\bar{k} = \vec{B} \text{ (say)}$$

$$\therefore |\vec{A}| = \sqrt{(4)^2 + (1)^2 + (-4)^2}$$

$$\Rightarrow |\vec{A}| = \sqrt{16+1+16}$$



$$\Rightarrow |\vec{A}| = \sqrt{33}$$

$$\therefore |\vec{B}| = \sqrt{(3)^2 + (3)^2 + (6)^2}$$

$$\Rightarrow |\vec{B}| = \sqrt{9 + 9 + 36}$$

$$\Rightarrow |\vec{B}| = \sqrt{54}$$

$$\Rightarrow |\vec{B}| = 3\sqrt{6}$$

$$\therefore \cos \theta = \frac{\vec{A} \cdot \vec{B}}{|\vec{A}| \cdot |\vec{B}|}$$

$$\Rightarrow \cos \theta = \frac{(\vec{i} + 4\vec{j} - 4\vec{k}) \cdot (3\vec{i} + 3\vec{j} + 6\vec{k})}{\sqrt{33} \times 3\sqrt{6}}$$

$$\Rightarrow \cos \theta = \frac{3 + 12 - 24}{3\sqrt{198}}$$

$$\Rightarrow \cos \theta = \frac{-9}{3\sqrt{198}}$$

$$\Rightarrow \theta = \cos^{-1} \left(\frac{-3}{\sqrt{198}} \right)$$

5. If $\vec{F} = \nabla(xy^3z^2)$ (or) $\vec{F} = \text{grad}(xy^3z^2)$, find $\text{div } \vec{F}$, and $\text{Curl } \vec{F}$ at the point $(1, -1, 1)$.

Sol:- $\vec{F} = \nabla(xy^3z^2)$

$$\Rightarrow \vec{F} = \nabla \phi$$

$$\Rightarrow \vec{F} = \frac{\partial \phi}{\partial x} \vec{i} + \frac{\partial \phi}{\partial y} \vec{j} + \frac{\partial \phi}{\partial z} \vec{k}$$

$$\Rightarrow \vec{F} = y^3z^2\vec{i} + 3xy^2z^2\vec{j} + 2xy^3z\vec{k}$$

$$i) \operatorname{div} \vec{F} = \nabla \cdot \vec{F}$$

$$\Rightarrow \operatorname{div} \vec{F} = \left(\frac{\partial}{\partial x} \vec{i} + \frac{\partial}{\partial y} \vec{j} + \frac{\partial}{\partial z} \vec{k} \right) \cdot (y^3 z^2 \vec{i} + 3xy^2 z^2 \vec{j} + 2xy^3 z \vec{k})$$

$$\Rightarrow \operatorname{div} \vec{F} = \frac{\partial}{\partial x} (y^3 z^2) + \frac{\partial}{\partial y} (3xy^2 z^2) + \frac{\partial}{\partial z} (2xy^3 z)$$

$$\Rightarrow \operatorname{div} \vec{F} = 0 + 6xy z^2 + 2xy^3$$

$$\Rightarrow \operatorname{div} \vec{F} = 6xy z^2 + 2xy^3$$

$$\Rightarrow \operatorname{div} \vec{F}_{P(1, -1, 1)} = 6(1)(-1)(1)^2 + 2(1)(-1)^3$$

$$\Rightarrow \operatorname{div} \vec{F}_{P(1, -1, 1)} = -6 - 2$$

$$\Rightarrow \operatorname{div} \vec{F} = -8$$

$$ii) \operatorname{Curl} \vec{F} = \nabla \times \vec{F}$$

$$\operatorname{Curl} \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^3 z^2 & 3xy^2 z^2 & 2xy^3 z \end{vmatrix}$$

$$\Rightarrow \operatorname{Curl} \vec{F} = (6xy^2 z - 6xy^2 z) \vec{i} - (2y^3 z - y^3 2z) \vec{j} + (3y^2 z^2 - 3y^2 z^2) \vec{k}$$

$$\Rightarrow \operatorname{Curl} \vec{F} = \left[(6(1)(-1)^2(1) - 6(1)(-1)^2(1)) \vec{i} \right] - \left[2(-1)^3(1) - (-1)^3 2(1) \right] \vec{j} + \left[3(-1)^2(1) - 3(-1)^2(1) \right] \vec{k}$$

$$\Rightarrow \operatorname{Curl} \vec{F} = [6-6] \vec{i} - [2-2] \vec{j} + [3-3] \vec{k}$$

$$\Rightarrow \operatorname{Curl} \vec{F} = 0 \vec{i} - 0 \vec{j} + 0 \vec{k}$$

$$\Rightarrow \operatorname{Curl} \vec{F} = \vec{0} \text{ is irrotational.}$$

6. If $\vec{F} = \nabla (x^3 + y^3 + z^3 - 3xyz)$ find $\text{div } \vec{F}$ & $\text{Curl } \vec{F}$.

Sol:- Let $\phi = x^3 + y^3 + z^3 - 3xyz$.

$$\therefore \vec{F} = \nabla \phi$$

$$\Rightarrow \vec{F} = \frac{\partial \phi}{\partial x} \vec{i} + \frac{\partial \phi}{\partial y} \vec{j} + \frac{\partial \phi}{\partial z} \vec{k}$$

$$\Rightarrow \vec{F} = (3x^2 - 3yz) \vec{i} + (3y^2 - 3zx) \vec{j} + (3z^2 - 3xy) \vec{k}$$

$$\Rightarrow \vec{F} = 3(x^2 - yz) \vec{i} + 3(y^2 - zx) \vec{j} + 3(z^2 - xy) \vec{k}$$

i) $\text{div } \vec{F}$:-

$$\text{WKT, } \nabla \cdot \vec{F} = \left(\frac{\partial}{\partial x} \vec{i} + \frac{\partial}{\partial y} \vec{j} + \frac{\partial}{\partial z} \vec{k} \right) \left(3(x^2 - yz) \vec{i} + 3(y^2 - zx) \vec{j} + 3(z^2 - xy) \vec{k} \right)$$

$$\Rightarrow \nabla \cdot \vec{F} = 3 \frac{\partial}{\partial x} (x^2 - yz) + 3 \frac{\partial}{\partial y} (y^2 - zx) + \frac{\partial}{\partial z} (z^2 - xy)$$

$$\Rightarrow \text{div } \vec{F} = 6x + 6y + 6z$$

$$\Rightarrow \text{div } \vec{F} = 6(x + y + z).$$

ii) $\text{Curl } \vec{F}$:-

$$\text{WKT, } \text{Curl } \vec{F} = \nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 3(x^2 - yz) & 3(y^2 - zx) & 3(z^2 - xy) \end{vmatrix}$$

$$\Rightarrow \text{Curl } \vec{F} = (-3x + 3x) \vec{i} - (-3y - 3y) \vec{j} + (-3z + 3z) \vec{k}$$

$$\Rightarrow \text{Curl } \vec{F} = 0 \vec{i} + 0 \vec{j} + 0 \vec{k} = \vec{0}$$

$$\Rightarrow \text{Curl } \vec{F} = \vec{0}$$

$\therefore \vec{F}$ is irrotational.

7. If $\vec{F} = x^2y\vec{i} + y^2z\vec{j} + z^2x\vec{k}$ then find $\text{curl}(\text{curl} \vec{F})$.

Sol:- $\text{Curl} \vec{F} = \nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2y & y^2z & z^2x \end{vmatrix}$

$$\Rightarrow \text{Curl} \vec{F} = (0 - y^2)\vec{i} - (z^2 - 0)\vec{j} + (0 - x^2)\vec{k}$$

$$\Rightarrow \text{Curl} \vec{F} = -y^2\vec{i} - z^2\vec{j} - x^2\vec{k} = \vec{A} \text{ (say)}$$

$$\therefore \text{Curl}(\text{Curl} \vec{F}) = \text{Curl} \vec{A}$$

$$= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ -y^2 & -z^2 & -x^2 \end{vmatrix}$$

$$\Rightarrow \text{Curl} \vec{A} = (0 + 2z)\vec{i} - (-2x - 0)\vec{j} + (0 + 2y)\vec{k}$$

$$\Rightarrow \text{Curl} \vec{A} = 2z\vec{i} + 2x\vec{j} + 2y\vec{k}$$

$$\Rightarrow \text{Curl}(\text{Curl} \vec{F}) = 2z\vec{i} + 2x\vec{j} + 2y\vec{k}$$

8. Show that $\vec{F} = \frac{x\vec{i} + y\vec{j}}{x^2 + y^2}$ is solenoidal and irrotational.

Sol:- $\vec{F} = \frac{x\vec{i} + y\vec{j}}{x^2 + y^2}$

$$\vec{F} = \frac{x\vec{i}}{x^2 + y^2} + \frac{y\vec{j}}{x^2 + y^2} + 0\vec{k}$$

$$i) \operatorname{div} \vec{F} = \nabla \cdot \vec{F}$$

$$\Rightarrow \operatorname{div} \vec{F} = \left[\frac{\partial}{\partial x} \vec{i} + \frac{\partial}{\partial y} \vec{j} + \frac{\partial}{\partial z} \vec{k} \right] \cdot \left[\frac{x}{x^2+y^2} \vec{i} + \frac{y}{x^2+y^2} \vec{j} + 0\vec{k} \right]$$

$$\Rightarrow \operatorname{div} \vec{F} = \frac{\partial}{\partial x} \left(\frac{x}{x^2+y^2} \right) + \frac{\partial}{\partial y} \left(\frac{y}{x^2+y^2} \right) + 0$$

$$\Rightarrow \operatorname{div} \vec{F} = \left| \frac{(x^2+y^2) - x(2x)}{(x^2+y^2)^2} \right| + \left| \frac{(x^2+y^2) - y(2y)}{(x^2+y^2)^2} \right|$$

$$\Rightarrow \operatorname{div} \vec{F} = \frac{(y^2 - x^2)}{(x^2+y^2)^2} + \frac{(x^2 - y^2)}{(x^2+y^2)^2}$$

$$\Rightarrow \operatorname{div} \vec{F} = \frac{y^2 - x^2 + x^2 - y^2}{(x^2+y^2)^2}$$

$$\Rightarrow \operatorname{div} \vec{F} = 0 \text{ is solenoidal.}$$

$$ii) \operatorname{Curl} \vec{F} = \nabla \times \vec{F}$$

$$\Rightarrow \operatorname{Curl} \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{x}{x^2+y^2} & \frac{y}{x^2+y^2} & 0 \end{vmatrix}$$

$$\Rightarrow \operatorname{Curl} \vec{F} = (0-0)\vec{i} - (0-0)\vec{j} + \left[\frac{-2xy}{(x^2+y^2)^2} + \frac{2xy}{(x^2+y^2)^2} \right] \vec{k}$$

$$\rightarrow \operatorname{Curl} \vec{F} = 0\vec{i} + 0\vec{j} + 0\vec{k} = \vec{0}$$

$$\Rightarrow \operatorname{Curl} \vec{F} \text{ is a irrotational.}$$

9. Find $\vec{F} = (x+3y)\vec{i} + (4-2z)\vec{j} + (x+az)\vec{k}$ is solenoidal.

Sol:- Given, $\vec{F}$ is solenoidal.

$$\Rightarrow \operatorname{div} \vec{F} = 0$$

$$\Rightarrow \nabla \cdot \vec{F} = 0$$

$$\Rightarrow \left(\frac{\partial}{\partial x} \vec{i} + \frac{\partial}{\partial y} \vec{j} + \frac{\partial}{\partial z} \vec{k} \right) \left((x+3y)\vec{i} + (4-2z)\vec{j} + (x+az)\vec{k} \right) = 0$$

$$\Rightarrow \frac{\partial}{\partial x} (x+3y) + \frac{\partial}{\partial y} (4-2z) + \frac{\partial}{\partial z} (x+az) = 0$$

$$\Rightarrow 1 + 1 + a = 0$$

$$\Rightarrow a + 2 = 0$$

$$\Rightarrow \boxed{a = -2}$$

10. Show that $\vec{F} = (y^2 - z^2 + 3yz - 2x)\vec{i} + (3xz + 2xy)\vec{j} + (3xy - 2xz + 2z)\vec{k}$ is both solenoidal and irrotational.

Sol:- $\therefore \operatorname{div} \vec{F} = \nabla \cdot \vec{F}$

$$\Rightarrow \operatorname{div} \vec{F} = \left(\frac{\partial}{\partial x} \vec{i} + \frac{\partial}{\partial y} \vec{j} + \frac{\partial}{\partial z} \vec{k} \right) \left((y^2 - z^2 + 3yz - 2x)\vec{i} + (3xz + 2xy)\vec{j} + (3xy - 2xz + 2z)\vec{k} \right)$$

$$\Rightarrow \operatorname{div} \vec{F} = \frac{\partial}{\partial x} (y^2 - z^2 + 3yz - 2x) + \frac{\partial}{\partial y} (3xz + 2xy) + \frac{\partial}{\partial z} (3xy - 2xz + 2z)$$

$$\Rightarrow \text{div } \vec{F} = -x + x - x + x$$

$$\Rightarrow \text{div } \vec{F} = 0$$

$\therefore \vec{F}$ is solenoidal.

$$\Rightarrow \text{Curl } \vec{F} = \nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ (y^2 - z^2 + 3yz - x) & (3xz + xy) & (3xy - xz + z) \end{vmatrix}$$

$$\Rightarrow \text{Curl } \vec{F} = (3x - 3x)\vec{i} + (3y - z + z - 3y)\vec{j} + (3z + y - y - 3z)\vec{k}$$

$$\Rightarrow \text{Curl } \vec{F} = 0\vec{i} + 0\vec{j} + 0\vec{k}$$

$$\Rightarrow \text{Curl } \vec{F} = \vec{0}$$

$\therefore \text{Curl } \vec{F}$ is irrotational

$\therefore \vec{F}$ is both solenoidal and irrotational.

11. Define an irrotational vector. Find the constants a, b, c such that $\vec{F} = (axy - z^3)\vec{i} + (bx^2 + z)\vec{j} + (bxz^2 + cy)\vec{k}$ is irrotational.

Sol:- Definition:-

Suppose $\vec{F} = F_1\vec{i} + F_2\vec{j} + F_3\vec{k}$ be a vector point function in x, y, z if $\text{curl } \vec{F} = \nabla \times \vec{F} = \vec{0}$, then $\vec{F}$ is called an "irrotational vector point function".

Given,

The vector point function, $\vec{F} = (axy - z^3)\vec{i} + (bx^2 + z)\vec{j} + (bxz^2 + cy)\vec{k}$ is irrotational.

$$\Rightarrow \text{curl } \vec{F} = \vec{0}$$

$$\Rightarrow \nabla \times \vec{F} = \vec{0}$$

$$\Rightarrow \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ ax - z^3 & bx^2 + z & bxz^2 + cy \end{vmatrix} = 0\vec{i} + 0\vec{j} + 0\vec{k}$$

$$\Rightarrow (c-1)\vec{i} - (bz^2 + 3z^2)\vec{j} + (2bx - ax)\vec{k} = 0\vec{i} + 0\vec{j} + 0\vec{k}$$

$$\Rightarrow (c-1)\vec{i} - (b+3)z^2\vec{j} + (2b-a)x\vec{k} = 0\vec{i} + 0\vec{j} + 0\vec{k}$$

$$\therefore c-1=0, \quad b+3=0, \quad 2b-a=0$$

$$\Rightarrow c=1, \quad \Rightarrow b=-3, \quad \Rightarrow 2b=a$$

$$\Rightarrow a = 2(-3) = -6$$

$$\therefore a = -6, b = -3, c = 1$$

12. Find the constants a, b, c such that $\vec{F} = (x+y+az)\vec{i} + (bx+2y-z)\vec{j} + (x+cy+2z)\vec{k}$ is irrotational, also find scalar potential ϕ for which $\vec{F} = \nabla \phi$.

Sol:- Given,

$$\vec{F} = (x+y+az)\vec{i} + (bx+2y-z)\vec{j} + (x+cy+2z)\vec{k} \rightarrow (1)$$

is irrotational.

$$\Rightarrow \text{curl } \vec{F} = \vec{0}$$

$$\Rightarrow \nabla \times \vec{F} = \vec{0}$$

$$\Rightarrow \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x+y+az & bx+2y-z & x+cy+2z \end{vmatrix} = 0\vec{i} + 0\vec{j} + 0\vec{k}$$

$$\Rightarrow (c+1)\bar{i} - (1-a)\bar{j} + (b-1)\bar{k} = 0\bar{i} + 0\bar{j} + 0\bar{k}$$

$$\Rightarrow c+1=0, \quad 1-a=0, \quad b-1=0$$

$$\Rightarrow c=-1, \quad a=1, \quad b=1$$

$$\therefore a=1, b=1, c=-1$$

$$\therefore (1) \Rightarrow \vec{F} = (x+y+z)\bar{i} + (x+2y-z)\bar{j} + (x-y+2z)\bar{k}$$

$$\text{and given } \vec{F} = \nabla \phi$$

$$(x+y+z)\bar{i} + (x+2y-z)\bar{j} + (x-y+2z)\bar{k} = \frac{\partial \phi}{\partial x} \bar{i} + \frac{\partial \phi}{\partial y} \bar{j} + \frac{\partial \phi}{\partial z} \bar{k}$$

$$\therefore \frac{\partial \phi}{\partial x} = x+y+z$$

$$\Rightarrow \partial \phi = (x+y+z) \partial x$$

$$\Rightarrow \int d\phi = \int (x+y+z) dx$$

$$\Rightarrow \phi = \frac{x^2}{2} + xy + xz + f_1(y, z) \rightarrow (2)$$

$$\therefore \frac{\partial \phi}{\partial y} = x+2y-z$$

$$\Rightarrow \partial \phi = (x+2y-z) \partial y$$

$$\Rightarrow \int d\phi = \int (x+2y-z) dy$$

$$\Rightarrow \phi = xy + y^2 - yz + f_2(x, z) \rightarrow (3)$$

$$\therefore \frac{\partial \phi}{\partial z} = x-y+2z$$

$$\Rightarrow \partial \phi = (x-y+2z) \partial z$$

$$\Rightarrow \int \partial \phi = \int (x-y+2z) \partial z$$

$$\Rightarrow \phi = xz - yz + z^2 + f_3(x, y) \rightarrow (4)$$

$$\Rightarrow \phi = \frac{x^2}{2} + y^2 + z^2 - yz + zx + xy$$

13. Find the constants a, b such that $\vec{F} = (axy + z^3)\vec{i} + (3x^2 - z)\vec{j} + (bxz^2 - y)\vec{k}$ is an irrotational, also find the scalar potential ϕ for which $\vec{F} = \nabla \phi$.

Sol:- Given,

$$\vec{F} = (axy + z^3)\vec{i} + (3x^2 - z)\vec{j} + (bxz^2 - y)\vec{k} \rightarrow (1)$$

is an irrotational.

$$\Rightarrow \text{curl } \vec{F} = \vec{0}$$

$$\Rightarrow \nabla \times \vec{F} = \vec{0}$$

$$\Rightarrow \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ (axy + z^3) & (3x^2 - z) & (bxz^2 - y) \end{vmatrix} = 0\vec{i} + 0\vec{j} + 0\vec{k}$$

$$\Rightarrow (-1+1)\vec{i} - (bz^2 - 3z^2)\vec{j} + (6x - ax)\vec{k} = 0\vec{i} + 0\vec{j} + 0\vec{k}$$

$$\Rightarrow 0\vec{i} + (3-b)z^2\vec{j} + (6-a)x\vec{k} = 0\vec{i} + 0\vec{j} + 0\vec{k}$$

$$\Rightarrow 3-b=0, \quad \Rightarrow 6-a=0$$

$$b=3 \quad \Rightarrow a=6$$

$$\therefore (1) \Rightarrow \vec{F} = (6xy + z^3)\vec{i} + (3x^2 - z)\vec{j} + (3xz^2 - y)\vec{k} \text{ and}$$

$$\vec{F} = \nabla \phi$$

$$\Rightarrow (6xy + z^3)\vec{i} + (3x^2 - z)\vec{j} + (3xz^2 - y)\vec{k} = \frac{\partial \phi}{\partial x}\vec{i} + \frac{\partial \phi}{\partial y}\vec{j} + \frac{\partial \phi}{\partial z}\vec{k}$$

$$\therefore \frac{\partial \phi}{\partial x} = 6xy + z^3$$

$$\Rightarrow \partial \phi = (6xy + z^3)\partial x$$

$$\Rightarrow \int \partial \phi = \int (6xy + z^3)\partial x$$

$$\Rightarrow \phi = 3x^2y + xz^3 + f_1(y, z) \rightarrow (2)$$

$$\therefore \frac{\partial \phi}{\partial y} = 3x^2 - z$$

$$\Rightarrow \partial \phi = (3x^2 - z) \partial y$$

$$\Rightarrow \int \partial \phi = \int (3x^2 - z) \partial y$$

$$\Rightarrow \phi = 3x^2y - yz + f_2(x, z) \rightarrow (3)$$

$$\therefore \frac{\partial \phi}{\partial z} = 3xz^2 - y$$

$$\Rightarrow \partial \phi = (3xz^2 - y) \partial z$$

$$\Rightarrow \int \partial \phi = \int (3xz^2 - y) \partial z$$

$$\Rightarrow \phi = xz^3 - yz + f_3(x, y) \rightarrow (4)$$

$$\therefore \phi = 3x^2y - yz + xz^3$$

14. Find the directional derivative of $\phi = x^2yz + 4xz^3$ at the point $(1, -2, 1)$ in the direction of vector $3\vec{i} - \vec{j} - 2\vec{k}$.

Sol:- Given, $\phi = x^2yz + 4xz^3$

$$\therefore \nabla \phi = \frac{\partial \phi}{\partial x} \vec{i} + \frac{\partial \phi}{\partial y} \vec{j} + \frac{\partial \phi}{\partial z} \vec{k}$$

$$\Rightarrow \nabla \phi = (2xyz + 4z^3) \vec{i} + x^2z \vec{j} + (x^2y + 8xz^2) \vec{k}$$

$$\therefore (\nabla \phi)_{P(1, -2, 1)} = \left[2(1)(-2)(1) + 4(1)^3 \right] \vec{i} + (1)^2(1) \vec{j} + \left[(1)^2(-2) + 8(1)(1) \right] \vec{k}$$

$$= 0\vec{i} + 0\vec{j} + 6\vec{k}$$

$$\therefore (\nabla \phi)_P = 0\vec{i} + 0\vec{j} + 6\vec{k}$$

and given,

$$\vec{d} = 3\vec{i} - \vec{j} - 3\vec{k}$$

$$\therefore \text{The normal } \hat{n} = \frac{\vec{d}}{|\vec{d}|} = \frac{3\vec{i} - \vec{j} - 3\vec{k}}{\sqrt{4+1+4}}$$

$$\Rightarrow \hat{n} = \frac{3\vec{i} - \vec{j} - 3\vec{k}}{3}$$

The required directional derivative,

$$D.D = \nabla \phi \cdot \hat{n}$$

$$\Rightarrow D.D = (0\vec{i} + \vec{j} + 6\vec{k}) \cdot \frac{3\vec{i} - \vec{j} - 3\vec{k}}{3}$$

$$\Rightarrow D.D = \frac{1}{3} | 0 + (-1) + (-18) |$$

$$\Rightarrow D.D = \frac{1}{3} (-1 - 18)$$

$$\Rightarrow D.D = \frac{-19}{3}$$

15. Find the directional derivative of $\phi = xy^3 + yz^3$ at $(3, -1, 1)$ in the direction of vector $\vec{i} + 3\vec{j} + 3\vec{k}$.

Sol:- Given,

$$\phi = xy^3 + yz^3$$

$$\therefore \nabla \phi = \frac{\partial \phi}{\partial x} \vec{i} + \frac{\partial \phi}{\partial y} \vec{j} + \frac{\partial \phi}{\partial z} \vec{k}$$

$$\Rightarrow (\nabla \phi) = (y^3)\vec{i} + (3xy^2 + z^3)\vec{j} + (3yz^2)\vec{k}$$

$$\therefore (\nabla \phi)_{P(3, -1, 1)} = (-1)^3 \vec{i} + (3(3)(-1)^2 + (1)^3)\vec{j} + (3(-1)(1)^2)\vec{k}$$
$$= -1\vec{i} + 7\vec{j} - 3\vec{k}$$

$$\therefore (\nabla \phi)_P = -\vec{i} + 7\vec{j} - 3\vec{k}$$

and given, $\vec{d} = \vec{i} + 2\vec{j} + 2\vec{k}$

$$\therefore \hat{n} = \frac{\vec{d}}{|\vec{d}|} = \frac{\vec{i} + 2\vec{j} + 2\vec{k}}{\sqrt{1+4+4}} = \frac{\vec{i} + 2\vec{j} + 2\vec{k}}{3}$$

The required directional derivative,

$$D.D = \nabla \phi \cdot \hat{n}$$

$$\Rightarrow D.D = (-\vec{i} + 7\vec{j} - 3\vec{k}) \cdot \frac{(\vec{i} + 2\vec{j} + 2\vec{k})}{3}$$

$$\Rightarrow D.D = \frac{1}{3}(-1 + 14 - 6)$$

$$\Rightarrow D.D = \frac{1}{3} \times 7$$

$$\Rightarrow D.D = \frac{7}{3}$$

16. Find the directional derivative, $\phi = 4xz^3 - 3x^2y^2z$ at $(2, -1, 2)$ along the direction vector $2\vec{i} - 3\vec{j} + 6\vec{k}$.

Sol:- Given,

$$\phi = 4xz^3 - 3x^2y^2z$$

$$\nabla \phi = \frac{\partial \phi}{\partial x} \vec{i} + \frac{\partial \phi}{\partial y} \vec{j} + \frac{\partial \phi}{\partial z} \vec{k}$$

$$\Rightarrow (\nabla \phi) = (4z^3 - 6xy^2z)\vec{i} + (-6x^2yz)\vec{j} + (12xz^2 - 3x^2y^2)\vec{k}$$

$$\Rightarrow (\nabla \phi)_{P(2, -1, 2)} = (4(2)^3 - 6(2)(-1)^2(2))\vec{i} + (-6(2)^2(-1)(2))\vec{j} + (12(2)(2)^2 - 3(2)^2(-1)^2)\vec{k}$$

$$\Rightarrow (\nabla \phi)_P = (32 - 24)\vec{i} + 48\vec{j} + (96 - 12)\vec{k}$$

$$\Rightarrow (\nabla \phi)_P = 8\vec{i} + 48\vec{j} + 84\vec{k}$$

Given, $\vec{d} = 8\vec{i} - 3\vec{j} + 6\vec{k}$

$$\hat{n} = \frac{\vec{d}}{|\vec{d}|} = \frac{8\vec{i} - 3\vec{j} + 6\vec{k}}{\sqrt{4+9+36}} = \frac{8\vec{i} - 3\vec{j} + 6\vec{k}}{7}$$

Directional derivatives = $\nabla \phi \cdot \hat{n}$

$$\Rightarrow D.D = 8\vec{i} + 48\vec{j} + 84\vec{k} \cdot \frac{(8\vec{i} - 3\vec{j} + 6\vec{k})}{7}$$

$$\Rightarrow D.D = \frac{16 - 144 + 504}{7}$$

$$\Rightarrow D.D = \frac{376}{7}$$

Vector integration:-

Line integrals:- The evaluation of an integral along with the given curve 'c' is called "The line integrals."

Let $\vec{F} = F_1\vec{i} + F_2\vec{j} + F_3\vec{k}$ be vector point function along the curve 'c', such that the position vector,

$$\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$$

$$\Rightarrow d\vec{r} = dx\vec{i} + dy\vec{j} + dz\vec{k}, \text{ then}$$

$$\Rightarrow \vec{F} \cdot d\vec{r} = (F_1\vec{i} + F_2\vec{j} + F_3\vec{k}) \cdot (dx\vec{i} + dy\vec{j} + dz\vec{k})$$

$$\Rightarrow \vec{F} \cdot d\vec{r} = F_1 dx + F_2 dy + F_3 dz$$

$$\Rightarrow \int_C \vec{F} \cdot d\vec{r} = \int_C (F_1 dx + F_2 dy + F_3 dz) \text{ is the line integral about the curve 'c'}$$

1. If $\vec{F} = 3xy\vec{i} - y^2\vec{j}$, evaluate $\int_C \vec{F} \cdot d\vec{r}$, where the 'C' is the variable of $y = 2x^2$ from the point (0,0) to (1,2).

Sol:- Given, $\vec{F} = 3xy\vec{i} - y^2\vec{j}$

WKT, $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$

$$\Rightarrow d\vec{r} = dx\vec{i} + dy\vec{j} + dz\vec{k}$$

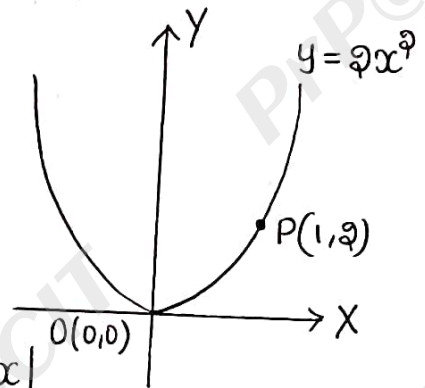
$$\therefore \vec{F} \cdot d\vec{r} = (3xy\vec{i} - y^2\vec{j}) \cdot (dx\vec{i} + dy\vec{j} + dz\vec{k})$$

$$\Rightarrow \vec{F} \cdot d\vec{r} = 3xy dx - y^2 dy$$

$$\therefore \int_C \vec{F} \cdot d\vec{r} = \int_C (3xy dx - y^2 dy) \longrightarrow (1)$$

where, C is $y = 2x^2$

$$\Rightarrow dy = 4x dx$$



$$\therefore (1) \Rightarrow \int_C \vec{F} \cdot d\vec{r} = \int_{x=0}^1 [3x(2x^2) dx - (2x^2)^2 4x dx]$$

$$= \int_{x=0}^1 (6x^3 dx - 16x^5 dx)$$

$$= \int_{x=0}^1 (6x^3 - 16x^5) dx$$

$$= 6 \left| \frac{x^4}{4} \right|_0^1 - 16 \left| \frac{x^6}{6} \right|_0^1$$

$$= 6 \left| \frac{1}{4} \right| - 16 \left| \frac{1}{6} \right|$$

$$= \frac{3}{2} - \frac{8}{3} \Rightarrow \frac{9-16}{6} = -\frac{7}{6}$$

Q. If $\vec{F} = x^2\vec{i} + xy\vec{j}$, evaluate $\int_C \vec{F} \cdot d\vec{r}$, which is from $(0,0)$ to $(1,1)$ along with the curve $y=x$, $y=\sqrt{x}$.

Sol:- Given, $\vec{F} = x^2\vec{i} + xy\vec{j}$

$$\therefore \text{WKT, } \vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$$

$$\Rightarrow d\vec{r} = dx\vec{i} + dy\vec{j} + dz\vec{k}$$

$$\Rightarrow \vec{F} \cdot d\vec{r} = x^2 dx + xy dy$$

$$\therefore \int_C \vec{F} \cdot d\vec{r} = \int_C (x^2 dx + xy dy) \rightarrow (1)$$

i) where C is a straight line $y=x$

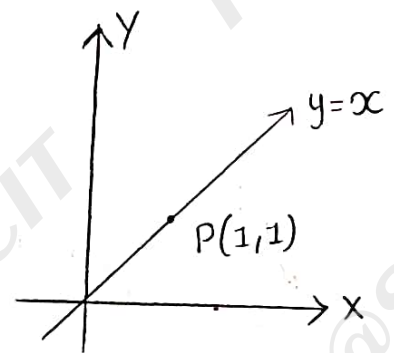
$\Rightarrow dy = dx$ and C is from $(0,0)$ to $(1,1)$

$$\therefore (1) \Rightarrow \int_C \vec{F} \cdot d\vec{r} = \int_{x=0}^1 (x^2 dx + x \cdot x dx)$$

$$= \int_{x=0}^1 (x^2 + x^2) dx$$

$$= 2 \int_0^1 x^2 dx$$

$$\int_C \vec{F} \cdot d\vec{r} = 2 \left| \frac{x^3}{3} \right|_0^1 = 2 \left| \frac{1}{3} \right| = \frac{2}{3}$$

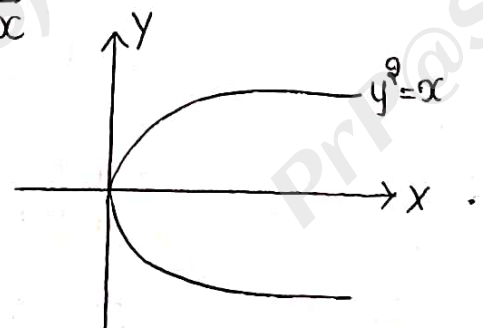


ii) where C is a parabola, $y=\sqrt{x}$

$$\Rightarrow y^2 = x$$

$$\Rightarrow 2y dy = dx$$

$$\therefore \int_C \vec{F} \cdot d\vec{r} = \int (y^4 2y dy + y^2 \cdot y dy)$$



$$= 9 \int_{y=0}^1 y^5 dy + \int_{y=0}^1 y^3 dy$$

$$= 9 \left| \frac{y^6}{6} \right|_0^1 + \left| \frac{y^4}{4} \right|_0^1$$

$$= 9 \left(\frac{1}{6} \right) + \left(\frac{1}{4} \right) \Rightarrow \frac{1}{3} + \frac{1}{4} = \frac{4+3}{12} = \frac{7}{12}$$

3. If $\vec{F} = (5xy - 6x^2)\vec{i} + (2y - 4x)\vec{j}$, evaluate $\int_C \vec{F} \cdot d\vec{r}$ along the curve $C: y = x^3$ in the xy plane from the point $(1,1)$ to $(2,8)$

Sol:- Given, $\vec{F} = (5xy - 6x^2)\vec{i} + (2y - 4x)\vec{j}$

$\therefore$ WKT, $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$

$$d\vec{r} = dx\vec{i} + dy\vec{j} + dz\vec{k}$$

$$\Rightarrow \vec{F} \cdot d\vec{r} = (5xy - 6x^2)dx + (2y - 4x)dy$$

$$\therefore \int_C \vec{F} \cdot d\vec{r} = \int_C (5xy - 6x^2)dx + (2y - 4x)dy \rightarrow (1)$$

where C is $y = x^3$.

$$dy = 3x^2 dx$$

$$\therefore (1) \Rightarrow \int_C \vec{F} \cdot d\vec{r} = \int_1^2 (5x \cdot x^3 - 6x^2)dx + (2x^3 - 4x)3x^2 dx$$

$$= \int_1^2 (5x^4 - 6x^2)dx + (6x^5 - 12x^3)dx$$

$$= \int_1^2 (5x^4 - 6x^2) + (6x^5 - 12x^3)dx$$

$$= \int_1^9 (5x^4 - 6x^3 + 6x^5 - 12x^3) dx$$

$$= \left[\frac{5x^5}{5} - \frac{6x^4}{4} + \frac{6x^6}{6} - \frac{12x^4}{4} \right]_1^9$$

$$= ((9)^5 - (1)) - 3((9)^3 - 1) + ((9)^6 - 1) - 3((9)^4 - 1)$$

$$= 31 - 14 + 63 - 45$$

$$\int_C \vec{F} \cdot d\vec{r} = 35$$

Green's theorem:-

Let $M(x, y)$, $N(x, y)$ be the two real value continuous function in x and y defined in the closed curve 'C', such that for the continuous partial derivatives $\frac{\partial M}{\partial y}$, $\frac{\partial N}{\partial x}$ then,

$$\oint_C M(x, y) dx + N(x, y) dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

1. Evaluate $\int_C (3x^2 - 8y^2) dx + (4y - 6xy) dy$ where the 'C' is a boundary of the region $y = \sqrt{x}$, $y = x^2$.

Sol:- Given,

$$\oint_C M dx + N dy = \int_C (3x^2 - 8y^2) dx + (4y - 6xy) dy$$

$$\therefore M = 3x^2 - 8y^2, \quad N = 4y - 6xy$$

$$\therefore \frac{\partial M}{\partial y} = -16y, \quad \frac{\partial N}{\partial x} = -6y$$

and given the curve C is bounded by the curves

$$y = \sqrt{x} \Rightarrow y^2 = x, \quad y = x^2$$

$\therefore$ By the green's theorem, WKT,

$$\int_C M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

$$= \iint_R (-6y + 16y) dx dy$$

$$= \iint_R 10y dx dy$$

$$= 10 \int_{x=0}^1 \int_{y=\sqrt{x}}^{x^2} y dy dx$$

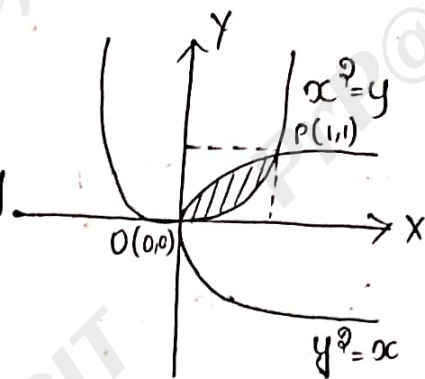
$$= 10 \int_0^1 \left[\frac{y^2}{2} \right]_{\sqrt{x}}^{x^2} dx$$

$$= 5 \int_0^1 (x^4 - x) dx$$

$$= 5 \left[\frac{x^5}{5} - \frac{x^2}{2} \right]_0^1$$

$$= 5 \left[\frac{-3}{10} \right]$$

$$\int_C M dx + N dy = \frac{-3}{2}$$



2. Evaluate $\int_C (xy + y^2) dx + x^2 dy$, where 'C' is of closed curve of the region bounded by $y = x$ & $y = x^2$.

Sol:-

Given,

$$\oint_C M dx + N dy = \int_C (xy + y^2) dx + x^2 dy$$

$$\therefore M = xy + y^2, N = x^2$$

$$\therefore \frac{\partial M}{\partial y} = x + 2y, \quad \frac{\partial N}{\partial x} = 2x$$

and given the curve 'C' is bounded by $y = x$ & $y = x^2$.

$\therefore$ By the Green's theorem,

WKT,

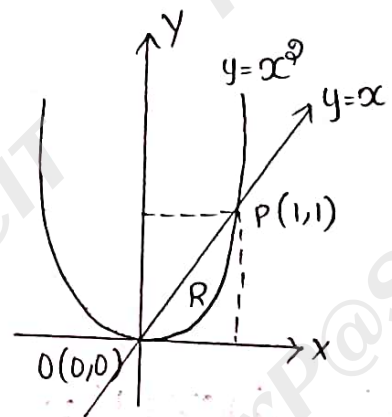
$$\int_C M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} + \frac{\partial M}{\partial y} \right) dx dy$$

$$= \iint_R (2x - x - 2y) dx dy$$

$$= \iint_R (x - 2y) dx dy$$

$$= \int_{x=0}^1 \int_{y=x^2}^x (x - 2y) dy dx$$

$$= \int_{x=0}^1 \left[xy - y^2 \right]_{y=x^2}^x dx$$



$$= \int_{x=0}^1 |(x^2 - x^3) - (x^3 - x^4)| dx$$

$$\Rightarrow \int_{x=0}^1 (x^4 - x^3) dx$$

$$\Rightarrow \int_c Mdx + Ndy = \left| \frac{x^5}{5} - \frac{x^4}{4} \right|_0^1$$

$$\Rightarrow \int_c Mdx + Ndy = \left| \frac{1}{5} - \frac{1}{4} \right| = \frac{4-5}{20}$$

$$\Rightarrow \int_c Mdx + Ndy = -\frac{1}{20}$$

3. Apply green's theorem $\int (3x - 8y^2) dx + (4y - 6xy) dy$ where 'c' is the boundary of the region $x=0$, $y=0$, $x+y=1$.

Sol:- Given,

$$\int Mdx + Ndy = \int_c (3x - 8y^2) dx + (4y - 6xy) dy$$

$$\therefore M = 3x - 8y^2, \quad N = 4y - 6xy$$

$$\therefore \frac{\partial M}{\partial y} = -16y, \quad \frac{\partial N}{\partial x} = -6y$$

and given the curve 'c' is bounded by $x=0$, $y=0$, $x+y=1$,

$$\therefore \text{WKT, } \int_c Mdx + Ndy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

$$\Rightarrow \iint_R (-6y + 16y) dx dy$$

$$= \iint_R 10y dx dy$$

$$= 10 \int_{x=0}^1 \int_{y=0}^{1-x} y dy dx$$

$$= 10 \int_{x=0}^1 \left| \frac{y^2}{2} \right|_0^{1-x} dx$$

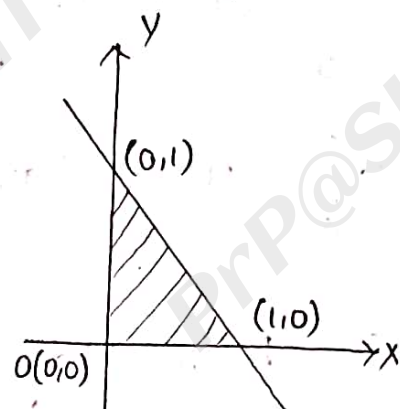
$$= 5 \int_0^1 (1-x)^2 dx$$

$$= 5 \int_0^1 (x-1)^2 dx$$

$$\Rightarrow \int_C M dx + N dy = 5 \left| \frac{(x-1)^3}{3} \right|_0^1$$

$$\Rightarrow \int_C M dx + N dy = \frac{5}{3} |0 - (-1)|$$

$$\Rightarrow \int_C M dx + N dy = \frac{5}{3}$$



Stokes theorem:-

Let $\vec{F} = F_1\vec{i} + F_2\vec{j} + F_3\vec{k}$ be a vector point function defined in an open surface 'L' bounded by a closed curve 'C' then the

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S \text{curl } \vec{F} \cdot \hat{n} \, ds$$

where $\hat{n}$ is the normal to the surface S and

$$\hat{n} \cdot ds = dydz\vec{i} + dzdx\vec{j} + dxdy\vec{k}.$$

1. Apply Stokes theorem to evaluate $\int_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = y^2\vec{i} + x^2\vec{j} - (x+z)\vec{k}$ and C is the boundary of the triangle with the vertices $(0,0,0)$, $(1,0,0)$, $(1,1,0)$.

Sol:- Given,

$$\vec{F} = y^2\vec{i} + x^2\vec{j} - (x+z)\vec{k}$$

$$\therefore \text{Curl } \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2 & x^2 & -(x+z) \end{vmatrix}$$

$$\Rightarrow \text{Curl } \vec{F} = (0-0)\vec{i} - (-1-0)\vec{j} + (2x-2y)\vec{k}$$

$$\Rightarrow \text{Curl } \vec{F} = \vec{j} + 2(x-y)\vec{k}$$

$\therefore$ The normal of the surface $\hat{n} = \vec{k}$

$$\therefore \text{Curl } \vec{F} \cdot \hat{n} = |\vec{j} + 2(x-y)\vec{k}| \cdot \vec{k}$$

$$\Rightarrow \text{Curl } \vec{F} \cdot \hat{n} = 2(x-y)$$

$\therefore$ By the Stokes theorem,

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S \text{Curl } \vec{F} \cdot \hat{n} \, ds$$

$$\Rightarrow \oint_C \vec{F} \cdot d\vec{r} = \iint_S 2(x-y) \, ds$$

$$\Rightarrow \oint_C \vec{F} \cdot d\vec{r} = 2 \int_{x=0}^1 \int_{y=0}^x (x-y) \, dx \, dy$$

$$\Rightarrow \oint_C \vec{F} \cdot d\vec{r} = 2 \int_{x=0}^1 \left(xy - \frac{y^2}{2} \right)_{y=0}^x \, dx$$

$$\Rightarrow \oint_C \vec{F} \cdot d\vec{r} = 2 \int_{x=0}^1 \left(x^2 - \frac{x^2}{2} \right) \, dx$$

$$\Rightarrow \oint_C \vec{F} \cdot d\vec{r} = 2 \int_0^1 \frac{x^2}{2} \, dx$$

$$\Rightarrow \oint_C \vec{F} \cdot d\vec{r} = \left| \frac{x^3}{3} \right|_0^1$$

$$\Rightarrow \oint_C \vec{F} \cdot d\vec{r} = \frac{1}{3}$$

Q. Apply Stokes theorem to evaluate,

$\vec{F} = (2x-y)\vec{i} - yz^2\vec{j} - y^2z\vec{k}$, where S is the upper half surface of the sphere $x^2+y^2+z^2=1$ and C is the boundary.

Sol:- Given,

$$\vec{F} = (2x-y)\vec{i} - yz^2\vec{j} - y^2z\vec{k}$$

Let

$$\nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ (2x-y) & -yz^2 & -y^2z \end{vmatrix} = \vec{k}$$

Taking the projection of the sphere $x^2+y^2+z^2=1$ on the XY -plane, we get the surface S as $x^2+y^2=1$ and $\hat{n} = \hat{k}$

By the definition of Stokes theorem, we have

$$\Rightarrow \oint_C \vec{F} \cdot d\vec{r} = \iint_S \text{Curl } \vec{F} \cdot \hat{n} \, ds$$

$$\Rightarrow \int_C \vec{F} \cdot d\vec{r} = \iint_S (\hat{k} \cdot \hat{k}) \, ds = \iint_S ds = S$$

= The area of the circle in XY -plane = π ,

(On S , the area = πr^2 , where $r=1$)

$$\therefore \oint_C \vec{F} \cdot d\vec{r} = \pi$$